Jargon Buster

AGP: Accelerated Graphics Port is a standard that defines the slot on which all the graphics cards are inserted on your motherboard. It allows the graphics cards to communicate with the other components in the computer such as memory, CPU; the data transfer rate on the AGP slot is higher than that of the PCI slot, thus enabling the graphic card to transfer massive textures faster to and from the main memory on the motherboard.

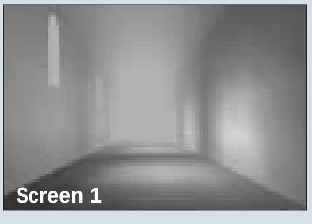
Blending: Reduction of two colour components to one component using interpolation

Alpha blending: Alpha is a fourth colour component (apart from RGB), that is used in 3D computer graphics to control the opacity of the rendered object. The transparent glass shown in this screenshot of *Quake III* is created using alpha blending.

Texels: A texel (Texture Element) is obtained by adding the texture data to the pixels.

Fill Rate: The fill rate is an indication of the amount of texels per second that a graphics card can transmit through its memory.

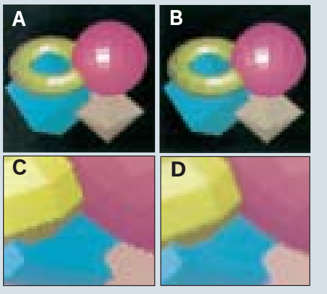
Texture mapping: A technique of mapping textures onto objects that are to be rendered on screen. Texture mapping is used to add realism to rendered scenes. For example in the screen shots alongside, Screen 1 is a plain rendered image without any texture mapping, whereas Screen 2 has the texture of a building facade applied to the walls.



Jitter: A pseudo-random displacement (shaking), of the objects in a scene, used in conjunction with the accumulation buffer to achieve special effects. For example, when you bang a car into another car, the effect that is produced is done using jitter.

Aliasing: Assigning individual colours to each pixel, regardless of whether it has to cover the complete pixel or only part of it. This results in jagged edges.

Anti-aliasing: In anti-aliasing, pixel colours are assigned based on the fraction of the pixel area that needs to be coloured. These lead to better and smoother edges. Picture A is an aliased image while picture B is an anti-aliased one. Picture C and Picture D are blown-up images of Picture A and B that clearly show the jagged edges of the aliased image and the blurred edges of the anti-aliased image.



MIP mapping: This is a texturing technique that creates sceneries that contain polygons at acute angles, disappearing into the distance. MIP mapping reduces the jagged effect from textures by mixing high and low resolution versions of the same texture.

Clipping: Eliminating the portions of the various geometric shapes in a rendered scene that lie outside the current viewing area.

Z buffer: Z buffer is that portion of the video memory that is reserved for holding depth information. This was not present in earlier 2D cards since they did not use any depth information. It is now a de facto standard in 3D cards.

Tesselation: Reducing a large surface into a mesh of small polygons.

Frame Buffer: The Frame Buffer is the portion of the video memory that stores all the display information before sending it to the screen.

Double buffering: New cards divide their frame buffer into two parts-a display buffer that stores the current frames display information, and a draw buffer that can be simultaneously used to draw the next frame.

High-end

This segment comprised the cream of the crop—the 5900 and the 9800 series graphics cards. It is for those who demand the very best in graphics cards, and do not shy away from dishing out a few bucks for a few FPS more. Here, money is not the priority, performance is. All the cards in this category are based on the ATI Radeon 9800 and nVidia 5900 chipsets. The ATI Radeon 9800 and its PRO versions yielded decent performances, but were way behind what the nVidia 5900 had to offer. The Gigabyte 9800 PRO (256 MB) logged the best scores amongst the 9800 series. The Hercules 3D Prophet 9800 PRO also performed well, but was overshadowed by the Club 3D ATi 9800 PRO, which performed better, and costs much less (Rs 28,000). The best performer in the ATI 9800 series comes with an unrealistic price tag, of Rs 44,000!

The Gainward GeForce FX 5900 Ultra is the best that your money can buy